

Gal Arnon

Postdoctoral Researcher · Bocconi University

gal.arnon@unibocconi.it
galarnon42.github.io

Research Interests

I am broadly interested in cryptography and complexity theory. My research focuses on probabilistic proof systems (e.g., PCPs and IOPs) as well as their cryptographic counterparts, with an emphasis on succinct non-interactive arguments. I study both the theoretical foundations and practical aspects of these systems.

Appointments

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| 2025–present | Postdoctoral Researcher, Bocconi University
Milan, Italy. Hosted by Prof. Alon Rosen. |
| 2025 | Research Fellow, Simons Institute for the Theory of Computing
Berkeley, USA. |

Education

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| 2020–2025 | Ph.D. Computer Science, Weizmann Institute of Science
Advisors: Prof. Moni Naor and Dr. Eylon Yogev.
Thesis: <i>New Advancements in Interactive Oracle Proofs: Theory, Practice, and Limitations.</i> |
| 2017–2020 | M.Sc. Computer Science, Weizmann Institute of Science
Advisor: Prof. Guy N. Rothblum.
Thesis: <i>On Prover-Efficient Public-Coin Emulation of Interactive Proofs.</i> |
| 2013–2017 | B.Sc. Electrical Engineering & Computer Science, Tel Aviv University
<i>Magna Cum Laude.</i> |

Awards

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| 2024 | Best Paper Award, CRYPTO 2024
For “STIR: Reed–Solomon Proximity Testing with Fewer Queries”. |
| 2024 | Esther Hellinger Memorial Prize for Academic Excellence, Weizmann Institute of Science |

Papers

Preprints

2. **Interactive Proofs for Batch Polynomial Evaluation**
Gal Arnon, Alessandro Chiesa, Giacomo Fenzi, and Eylon Yogev
Manuscript
1. **Pairing-Based SNARGs with Two Group Elements**
Gal Arnon, Jesko Dujmovic, and Eylon Yogev
Manuscript

Publications

11. **Designated-Verifier SNARGs with One Group Element**
Gal Arnon, Jesko Dujmovic, and Yuval Ishai
CRYPTO 2025
10. **Towards a White-Box Secure Fiat-Shamir Transformation**
Gal Arnon and Eylon Yogev
CRYPTO 2025, ZKProof 7.
9. **Instance Compression, Revisited**
Gal Arnon, Shany Ben-David, and Eylon Yogev
EUROCRYPT 2025
8. **WHIR: Reed–Solomon Proximity Testing with Super-Fast Verification**
Gal Arnon, Alessandro Chiesa, Giacomo Fenzi, and Eylon Yogev
EUROCRYPT 2025, ZKSummit 12, ZKProof 7.
7. **Hamming Weight Proofs of Proximity with One-Sided Error**
Gal Arnon, Shany Ben-David, and Eylon Yogev
TCC 2024
6. **STIR: Reed–Solomon Proximity Testing with Fewer Queries**
Gal Arnon, Alessandro Chiesa, Giacomo Fenzi, and Eylon Yogev
CRYPTO 2024, ZKSummit 11, ZKProof 6. **[Best Paper Award]**
5. **IOPs with Inverse Polynomial Soundness Error**
Gal Arnon, Alessandro Chiesa, and Eylon Yogev
FOCS 2023, ITC 2024 Highlights
4. **A Toolbox for Barriers on Interactive Oracle Proofs**
Gal Arnon, Amey Bhangale, Alessandro Chiesa, and Eylon Yogev
TCC 2022
3. **Hardness of Approximation for Stochastic Problems via Interactive Oracle Proofs**
Gal Arnon, Alessandro Chiesa, and Eylon Yogev
CCC 2022
2. **A PCP Theorem for Interactive Proofs and Applications**
Gal Arnon, Alessandro Chiesa, and Eylon Yogev
EUROCRYPT 2022

1. On Prover-Efficient Public-Coin Emulation of Interactive Proofs

Gal Arnon and Guy N. Rothblum

ITC 2021

Media Coverage

- Quanta Magazine [Computer Scientists Figure Out How to Prove Lies](#) mentions our work “Towards a White-Box Secure Fiat-Shamir Transformation”.
- ZK Podcast Episode 321 - [Interview on the STIR protocol](#).

Selected Talks

- **Fiat-Shamir Attacks and Mitigations**
 - Proofs Workshop, “Cryptography, 10 Years Later” Semester, Simons Institute, Berkeley, USA. *July 2025*.
 - Proofs Reading Group, “Cryptography, 10 Years Later” Semester, Simons Institute, Berkeley, USA. *June 2025*.
 - NTT Cryptography Seminar (Virtual). *April 2025*.
- **Instance Compression, Revisited**
 - Foundations Reading Group, “Cryptography, 10 Years Later” Semester, Simons Institute, Berkeley, USA. *July 2025*.
- **STIR: Reed–Solomon Proximity Testing with Fewer Queries**
 - CRYPTO 2024, Santa Barbara, USA. *August 2024*.
 - Interuniversity TCS Student Seminar, Tel Aviv University, Israel. *May 2024*.
 - Theory Lunch, Weizmann Institute of Science, Israel. *May 2024*.
 - HUJI TCS Seminar, Jerusalem, Israel. *May 2024*.
 - StarkWare Industries, Netanya, Israel. *April 2024*.
 - ZKSummit 11, Athens, Greece. *April 2024*.
- **IOPs with Inverse Polynomial Soundness Error**
 - ITC 2024 Highlights Track, Stanford University, USA. *August 2024*.
 - Technion TCS Seminar, Haifa, Israel. *February 2024*.
 - ZK Study Club (Virtual). *October 2023*.
 - StarkWare Industries, Netanya, Israel. *September 2023*.
 - Interuniversity TCS Student Seminar, Tel Aviv University, Israel. *July 2023*.
 - IST Austria TCS Seminar, Vienna, Austria. *June 2023*.
- **How To Be Convinced While Barely Listening (Even to Yourself)**
 - Theory Lunch, Weizmann Institute of Science, Israel. *May 2025*.
 - EPFL CS Theory Reading Group, Lausanne, Switzerland. *May 2023*.
 - Efficient Probabilistic Proofs Workshop, Bertinoro, Italy. *July 2022*.
- **A Toolbox for Barriers on Interactive Oracle Proofs**
 - TCC 2022, Chicago, USA. *November 2022*.
- **Hardness of Approximation for Stochastic Problems via Interactive Oracle Proofs**
 - CCC 2022, Philadelphia, USA (Virtual). *July 2022*.
- **A PCP Theorem for Interactive Proofs and Applications**
 - EUROCRYPT 2022, Trondheim, Norway. *May–June 2022*.

- Theory Lunch, Weizmann Institute of Science, Israel. *July 2021.*
- **On Prover-Efficient Public-Coin Emulation of Interactive Proofs**
 - ITC 2021 (Virtual). *July 2021.*
 - “Proofs, Consensus, and Decentralizing Society” Program Seminar, Simons Institute, Berkeley, USA. *October 2019.*

Service

Proximity Prize	Committee member and judge for the Proximity Prize initiative of the Ethereum Foundation — a bounty to prove or disprove Reed–Solomon proximity gap conjectures.
Workshop	Lattices Meet Hashes: Recent Advances in Post-Quantum Zero-Knowledge Proofs . Postdoctoral workshop, Bernoulli Center, EPFL, Lausanne. Co-organized with Ngoc Khanh Nguyen. May 2023.
Program Comm.	TCC 2026
Reviews	CCC (2024) · CRYPTO (2019, 2022, 2023, 2024, 2025, 2026) · EUROCRYPT (2023, 2026) · FOCS (2025) · ITC (2025) · ITCS (2022, 2026) · SICOMP (2026) · SODA (2024) · STOC (2025) · TCC (2021, 2023, 2025)

Teaching

July 2023	Teaching Assistant , <i>Foundations and Frontiers of Probabilistic Proofs</i> MSRI (SLMath) Summer Graduate School, ETH Zurich, Switzerland.
July–Aug 2021	Teaching Assistant , <i>Foundations and Frontiers of Probabilistic Proofs</i> MSRI (SLMath) Summer Graduate School, Virtual.
Sept 2018	Instructor , <i>Mini-Course on Zero-Knowledge Proofs</i> Amos de-Shalit Summer School, Weizmann Institute of Science.